SYSTEM ARCHITECTURE & PRODUCT SPECIFICATION BLUEPRINT

Multi-Tier Affiliate & Sub-Affiliate Tracking Platform

End-to-End Business Flow, Dynamic URL Parameter Registry, Traffic Analytics & UI Wireframes

Role: Master Affiliate Hub & Sub-Network | **Attribution Model:** Server-to-Server (S2S) Click ID Postback | **Channels:** Direct Links + Sub-Affiliate Tier

1. How the Business & Application Works (In Simple Words)

BUSINESS MODEL

Your web application acts as a **Master Affiliate Distribution Hub**. Instead of promoting a brand's products only by yourself, you sign a primary partnership with a Brand (or Affiliate Network) and build a portal where both **you (Direct)** and **registered Sub-Affiliates** (influencers, content creators, bloggers, and media buyers) generate trackable promotional links. Because customers complete their purchase on the Brand's external website, your application sits in the middle of every click—recording visitor traffic first, attaching a unique digital receipt number (`click_id`), and automatically splitting the commission when the Brand notifies your server of a completed sale.

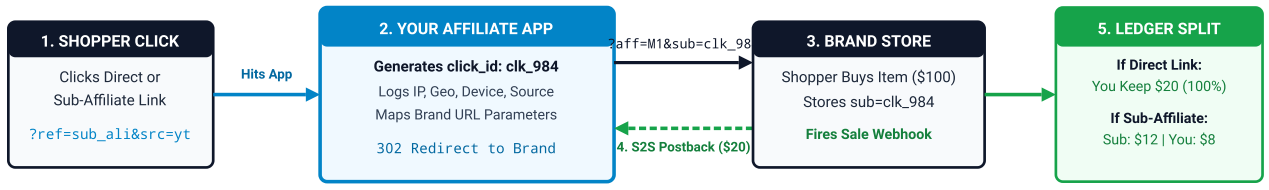


Figure 1.1: Complete lifecycle of a tracked click, parameter injection, brand checkout, S2S postback, and automated commission split.

2. Direct Promotion vs. Sub-Affiliate Economics

REVENUE STREAMS

Mode A: Direct Master Link Promotion

100% Margin

When you share a product link on your own website, social channels, or email list using `?ref=direct`, your system generates a `click_id` tied directly to your Master Admin account. When the customer buys a $100 item and the brand pays a 20% ($20) commission, **$20.00 is credited to your Net Master Profit** with $0.00 sub-affiliate liability.

Mode B: Sub-Affiliate Network Promotion

Override Margin

When a registered partner shares their link (e.g., `?ref=sub_ali_05`), your system links the generated `click_id` to their sub-affiliate ID. When the $20 commission arrives from the brand, your app automatically applies their agreed split (e.g., 60% of commission = **$12.00 to Sub-Affiliate Wallet**) and retains **$8.00 as your Master Override Profit**.

3. Step-by-Step Operational Flow (6 Stages)

SYSTEM WORKFLOW

1. **Step 1 — Brand & Parameter Setup (Admin):** You register a Brand in your dashboard, enter your Master Affiliate ID provided by that brand, configure their URL Parameter Template, and add promotional items (products/banners) to your catalog.
2. **Step 2 — Link Creation (Direct or Sub-Affiliate):** You or a logged-in sub-affiliate choose an item to promote. Your app builds a tracking URL pointing to your domain (e.g., `https://yourapp.com/go/shoe-99?ref=sub_104&src=tiktok&camp=eid_sale`).
3. **Step 3 — Traffic Capture & Click ID Creation:** When a visitor clicks the link, your server intercepts the request in <50ms, creates a unique `click_id` (e.g., `clk_88412`), saves the visitor's IP, city, device, browser, referrer, and `ref` code in your database, and looks up the target Brand's parameter rules.
4. **Step 4 — Dynamic Outbound Redirect:** Your server redirects the shopper to the Brand's product page with your Master ID and the newly created `click_id` injected into the Brand's required URL parameters.
5. **Step 5 — Purchase & Brand Postback Webhook:** The shopper buys the item on the Brand's website. The Brand's system fires an automated Server-to-Server (S2S) Postback HTTP request to your webhook URL containing the `click_id`, order ID, sale amount, and total commission earned.
6. **Step 6 — Attribution & Wallet Settlement:** Your webhook verifies the security token, matches the `click_id` to the exact click record, calculates the commission split between you and the sub-affiliate, and logs the sale as *Pending* until the brand's return window clears.

4. Registering Affiliate URL Parameter Definitions

PARAMETER REGISTRY

Different brands and networks (Impact, CJ, Shopify, custom carts) expect different URL query parameter names. Instead of hardcoding URLs, your application uses a **Two-Layer Parameter Definition Registry** that translates inbound traffic tags into outbound brand parameters automatically.

4.1 Inbound Tracking Parameters (Used on Your App's Promotional Links)

Parameter Key	System Role	Example Value	Definition & Attribution Purpose
`ref` / `aff_id`	Promoter Identifier	`direct` or `sub_ali_05`	Identifies whether the click belongs to your Direct link or a specific Sub-Affiliate account.
`src` / `utm_source`	Traffic Channel	`youtube`, `instagram`, `blog`	Tracks which platform or traffic source generated the visitor click for source reporting.
`camp` / `utm_campaign`	Campaign Tag	`summer_drop_26`, `review_vid`	Groups traffic and sales by marketing campaign, video, or seasonal promotion.
`sub1`, `sub2`	Custom Sub-Tags	`header_btn`, `story_link_2`	Optional free-form parameters allowing sub-affiliates to test specific ad placements.

4.2 Outbound Brand Parameter Definitions (Dynamic Translation Registry)

Brand / Network Type	Master ID Param	Click ID Param	Constructed Outbound Redirect URL Template
Custom Brand Store	`partner_id`	`click_ref`	`https://brand.com/item?partner_id={master_id}&click_ref={click_id}&sub={ref}`
Impact / CJ Network	*Path Segment*	`subId1`	`https://brand.pxf.io/c/{master_id}/4912/81?subId1={click_id}&subId2={ref}`
Shopify / Refersion	`rfsn`	`subid`	`https://shop.com/products/bag?rfsn={master_id}&subid={click_id}&utm_source=`

5. Core Database Architecture (Holding All Records)

DATA SCHEMA

To keep an auditable ledger of every direct click, sub-affiliate click, parameter rule, and sold item, your relational database (PostgreSQL / MySQL) requires five interconnected tables linked by foreign keys and indexed on `click_id` and `ref_code`.

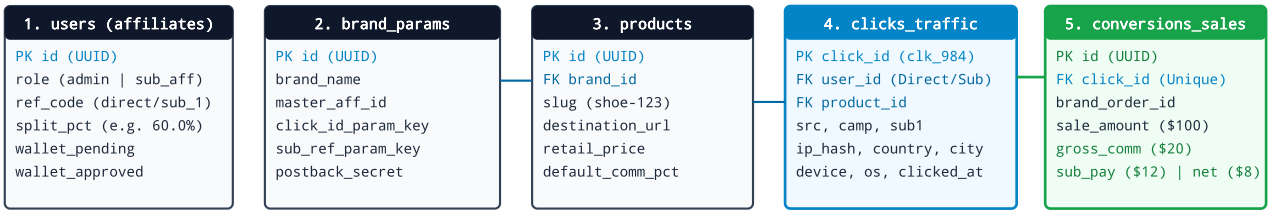


Figure 3.1: Core Entity-Relationship Schema connecting Sub-Affiliates, Brand Parameter Rules, Traffic Logs, and Split Sales Ledgers.

6. Website Traffic & Sales Reporting Suite

ANALYTICS ENGINE

Your reporting engine aggregates real-time data from `clicks_traffic` and `conversions_sales` across four core views so you can monitor website traffic quality, channel ROI, and sub-affiliate payouts:

1. Direct vs. Sub-Affiliate Revenue Report

Financial

Compares traffic and sales from your own direct links (`ref=direct`) against your sub-affiliate network. Displays **Gross Brand Commission**, **Sub-Affiliate Payout Liability**, and **Net Master Profit** by date range.

2. Sub-Affiliate Leaderboard & Fraud Audit

Partner Ops

Ranks every sub-affiliate by Unique Clicks, Orders Sold, **Conversion Rate (CR%)**, and **Earnings Per Click (EPC)**. Flags suspicious traffic spikes (e.g., 5,000 bot clicks with 0% conversion).

3. Traffic Source, Campaign & Geo Report

Traffic Analytics

Breaks down visitor sessions by `src` (YouTube, TikTok, Facebook, SEO), `camp` tag, Country/City, and Device Type (Mobile iOS/Android vs. Desktop) to show which channels bring real buyers.

4. Item / Product Conversion Funnel

Catalog ROI

Tracks each promoted item from **Landing Page Views → Outbound Brand Clicks (CTR%) → Confirmed Orders → Reversal/Refund Rate**, helping you highlight high-converting items to sub-affiliates.

Key Calculated Metrics to Display on Every Report Table

Click-Through Rate (CTR): `(Outbound Brand Clicks ÷ Product Page Views) × 100` | **Conversion Rate (CR%):** `(Confirmed Sales ÷ Unique Outbound Clicks) × 100` | **Earnings Per Click (EPC):** `Total Commission Earned ÷ Unique Outbound Clicks` | **Net Override Margin:** `Gross Brand Commission − Sub-Affiliate Payout`.

7. Application UI Wireframes (Master Admin & Sub-Affiliate Portal)

INTERFACE BLUEPRINT

The wireframe below illustrates the two primary screens of your web application: the **Master Admin Dashboard** (left, featuring KPI cards, Direct vs. Sub-Affiliate revenue bars, and the Parameter Registry modal) and the **Sub-Affiliate Portal** (right, featuring the custom Link Generator and personal sales ledger).

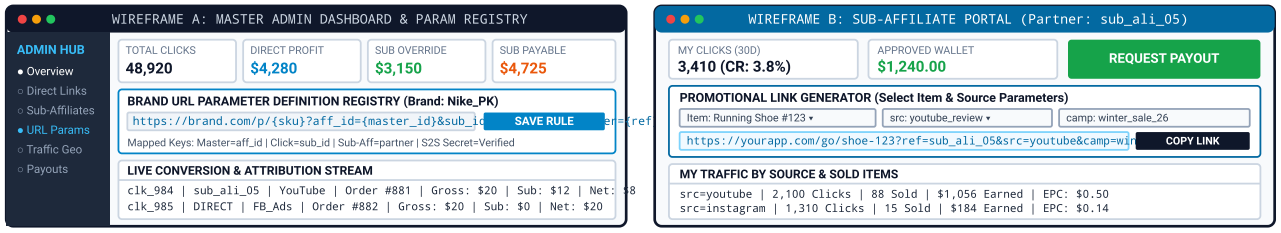


Figure 4.1: Wireframes of the Master Admin Command Center (left) and the Sub-Affiliate Partner Portal (right).

8. Core API Routes & Implementation Checklist

ENGINEERING SPEC

To launch the platform cleanly, your backend service exposes five primary endpoints and enforces strict role-based isolation so sub-affiliates can never view your master commission rates or other partners' traffic:

HTTP Endpoint	Access Scope	Execution Responsibility
`GET /go/:product_slug`	Public Shopper	Reads `ref`, `src`, `camp`; logs IP/Geo/Device; creates `click_id`; maps brand URL params; executes HTTP 302 redirect to Brand.
`GET|POST /api/v1/postback`	Brand Webhook	Validates brand secret key; matches `click_id`; prevents duplicate `order_id`; splits commission into Direct vs. Sub-Affiliate ledgers.
`POST /api/admin/brand-params`	Master Admin	Registers or updates outbound URL parameter templates (`{master_id}`, `{click_id}`, `{ref}`) and commission rates per brand.
`GET /api/reports/master`	Master Admin	Returns full traffic telemetry, Direct vs. Sub-Affiliate revenue comparison, sub-affiliate leaderboard, and geo/device breakdowns.
`GET /api/portal/my-stats`	Sub-Affiliate	Returns filtered clicks, conversion rates, channel breakdown, and wallet balances strictly scoped to the logged-in `sub_affiliate_id`.

Recommended Implementation Roadmap (4 Phases)

Phase 1 (Core Redirect & Parameter Engine): Build the `brand_params` registry and the `/go/:slug` redirector that generates `click_id` values and logs visitor telemetry. | **Phase 2 (Postback & Split Ledger):** Build the `/api/v1/postback` endpoint to capture sales and split Direct vs. Sub-Affiliate commissions. | **Phase 3 (Sub-Affiliate Portal):** Launch partner sign-up, link generator UI, and scoped earnings tables. | **Phase 4 (Advanced Traffic Analytics):** Add GeoIP lookup, device parsing, and automated payout approval workflows.